

One-Way Delay Estimation Over the Internet

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1. Introduction

One-Way Delay (OWD) refers to the time it takes for a packet to be transmitted across a network from source to destination.

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PING google.com (192.178.56.78) 56(84) bytes of data.  
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OWD is usually approximated as $\frac{RTT}{2}$ (many assumptions made).

2. Derivation

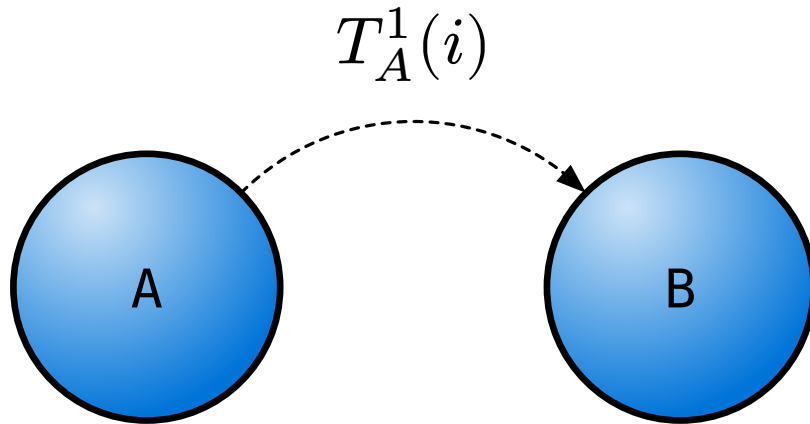
- *system clock*: $C(t)$,
- *absolute offset*: offset from a “true” clock, $C_{A(t)} - t$
- *relative offset*: offset from another clock, $\theta(t) = C_{A(t)} - C_{B(t)}$,
- *frequency*: rate of progression, $C'(t) = \frac{dC(t)}{dt}$,
- *clock ratio*: frequency ratio $\alpha = \frac{C_{A(t)}}{C_{B(t)}}$,
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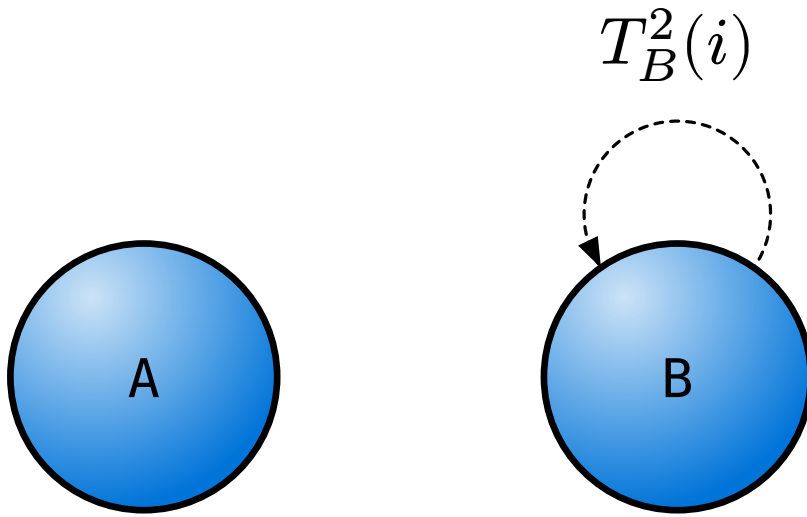
The relationship between two clocks is then:

$$C_{A(t)} = \theta(t) + \alpha C_{B(t)}.$$

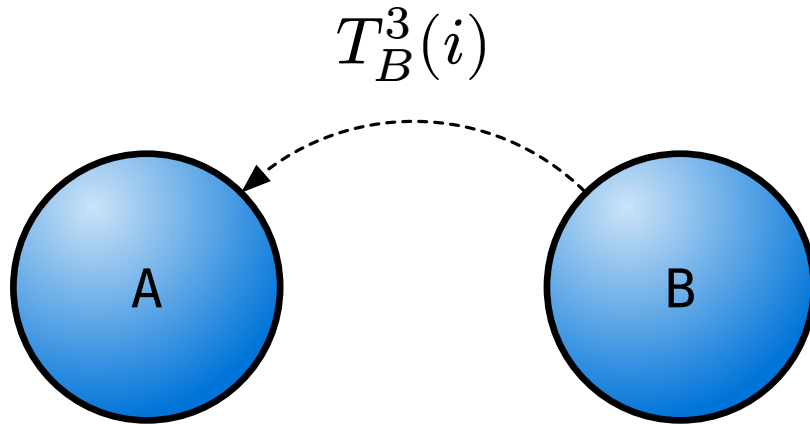
3. Clock Information Sharing



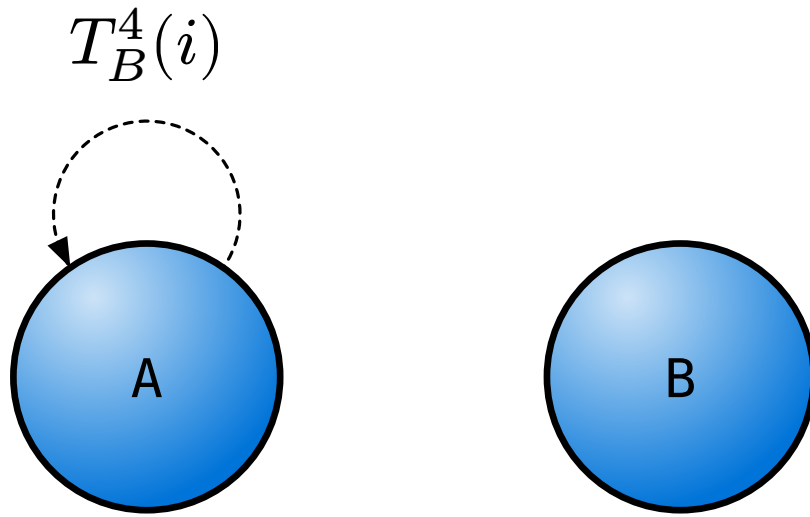
$T_A^1(i)$ — message i with timestamp.



$T_B^2(i)$ — Add a
timestamp of reception



$T_B^3(i)$ — Add a timestamp when sent



$T_A^4(i)$ — Add a timestamp when received

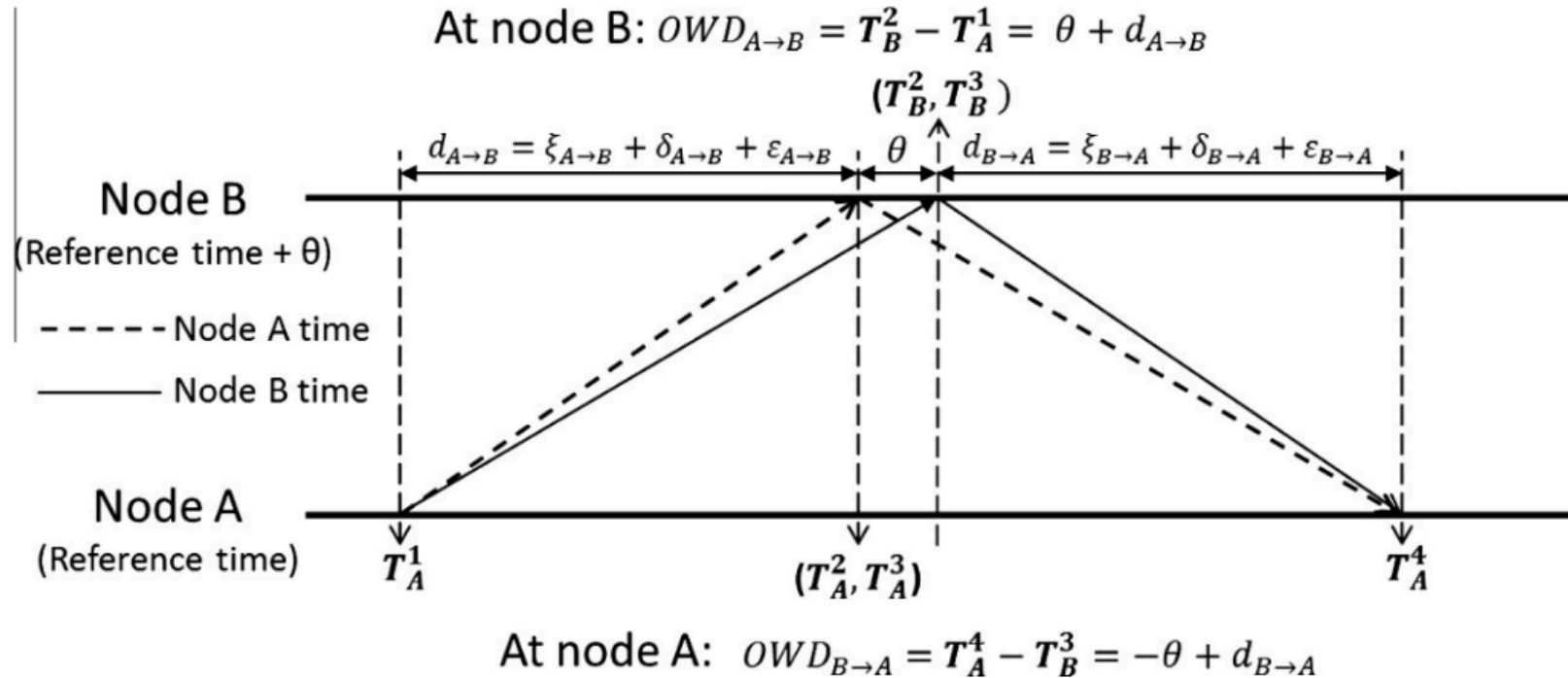


Fig. 1. Time stamp exchange between two nodes with nonzero offset θ . Node A represents the true time (dashed line).

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$$\text{OWD} = T_{\text{recv}} - T_{\text{sent}},$$

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$\alpha \approx 1$ for short time probes and small packet sizes.

4. Previous models

$$\hat{\theta}_{A \leftrightarrow B} = \min_{i \mid 0 < i \leq k} \left(\min_j \frac{\text{OWD}_{A \rightarrow B_k}(j) - \text{OWD}_{B_k \rightarrow A}(j)}{2} \right)$$

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$$\sum_{i=1}^n \nu_i (X_{(i)} - Y_{(i)})$$

$$\hat{\theta}_{A \leftrightarrow B} = \frac{1}{2} \left\{ \max_{1 \leq i \leq n} \{S_i - \alpha U_i\} + \min_{1 \leq i \leq n} \{T_i - \alpha V_i\} \right\}$$

5. Proposed model

- Independent one-way offset estimation
- Collaborative offset estimation

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In general, the distribution of δ cannot be succinctly characterized. It is affected by the network topology and its possible rearrangements.

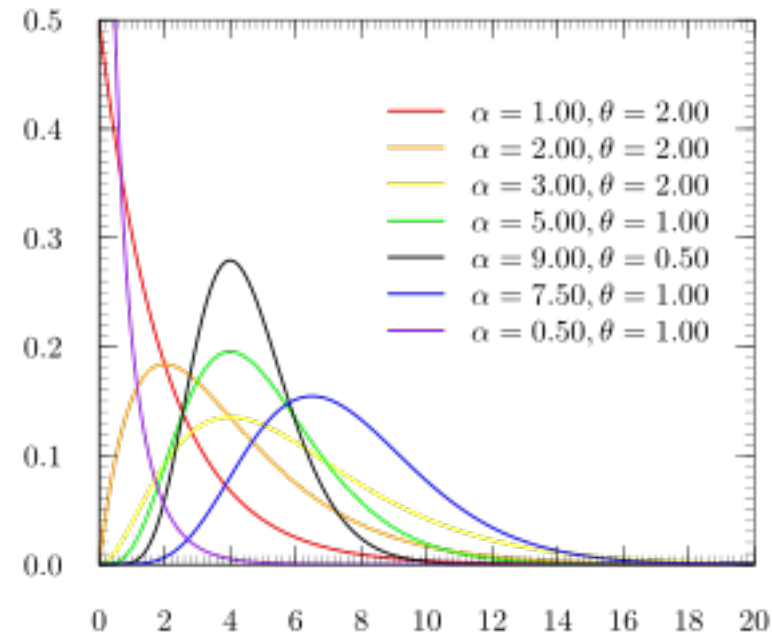
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In general, the distribution of δ cannot be succinctly characterized. It is affected by the network topology and its possible rearrangements. Over a fixed path, it can be *shown* that it follows a gamma distribution.



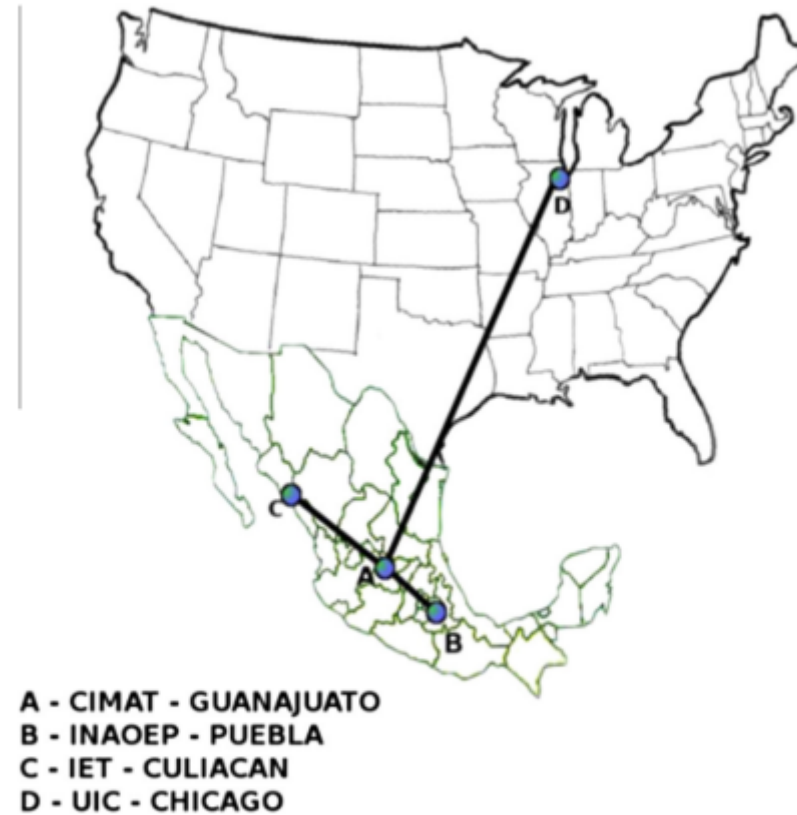


Fig. 2. Selected paths for the characterization of the one-way queuing delay distribution.

A week of experimentation for each link, 48-bytes UDP with $\{10, 20, 30, 40, 80, 160, 500, 1000\}ms$ inter-packet separations.

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A preprocess is done to remove offset and skewness.

A maximum likelihood estimate yielded

$$f(\chi, \rho, \beta, \gamma) = \frac{\left(\frac{\chi-\gamma}{\beta}\right)^{\rho-1} e^{-\frac{\chi-\gamma}{\beta}}}{\beta\Gamma(\rho)}, \quad (\rho > 0, \beta > 0; \chi > \gamma)$$

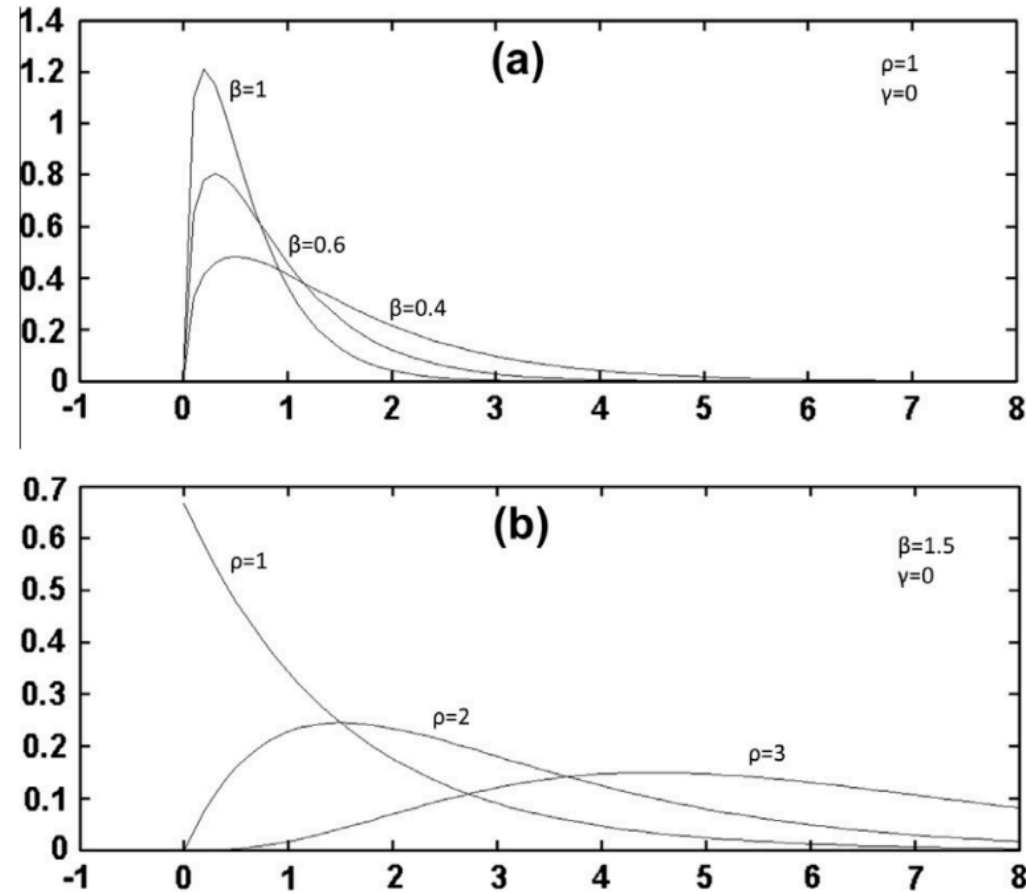


Fig. 3. Gamma distribution examples with variable scale parameter (a) and variable shape parameter (b).

Instead of incrementing the number of links (which was seen to have a similar distribution), only the traffic condition was considered.

- Use a predefined gamma distribution
- Assume that the queuing delays $\{\delta_i\}_{i=1}^n$ are i.i.d.
- A sends n 48-byte UDP packets at a rate of $\frac{1}{d}$ with timestamp info from sender.
- A prepares to receive n packets.
- B sends back the packets at the same rate $\frac{1}{d}$.
- B estimates the parameters of the gamma distribution.

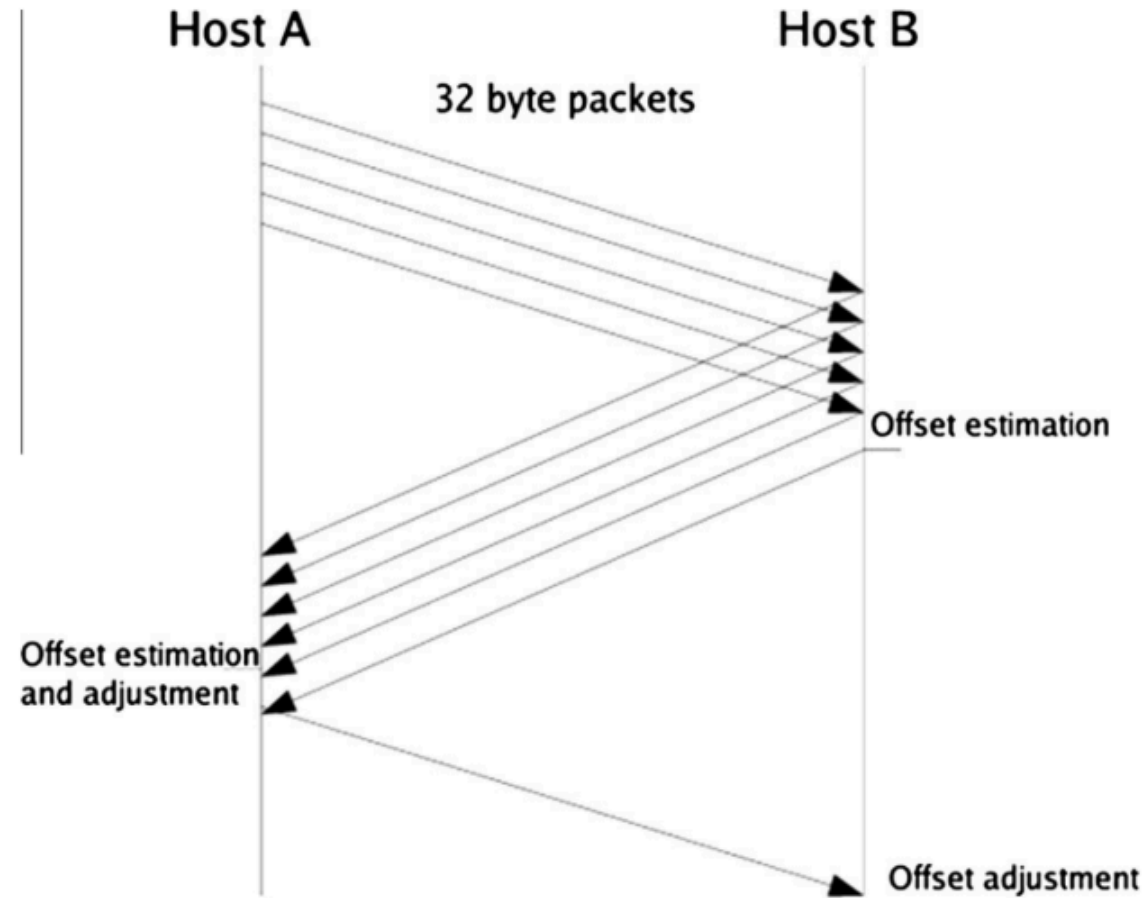


Fig. 5. Packet exchange policy for collaborative offset estimation.

$$\rho = \left(\frac{\overline{\chi}}{s} \right)^2,$$

$$\beta = \frac{s^2}{\overline{\chi}}$$

Let $x(i) = T_{\text{recv}(i)} - T_{\text{send}(i)}$, $\tilde{x}^{(j)} = \text{sort}_{j(x(i))}$, $y^{(i)}$ the theoretical ordered random values of the gamma distribution.

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Let $x(i) = T_{\text{recv}(i)} - T_{\text{send}(i)}$, $\tilde{x}^{(j)} = \text{sort}_{j(x(i))}$, $y^{(i)}$ the theoretical ordered random values of the gamma distribution. Now, we calculate either

$$\gamma = \text{offset} + \text{propagation delay} + \text{transmission delay}$$

or eliminate δ .

$$\hat{\theta} = \gamma = \theta + \xi$$

$$\widehat{\text{OWD}}_i = \tilde{x}^i - \hat{\theta}$$

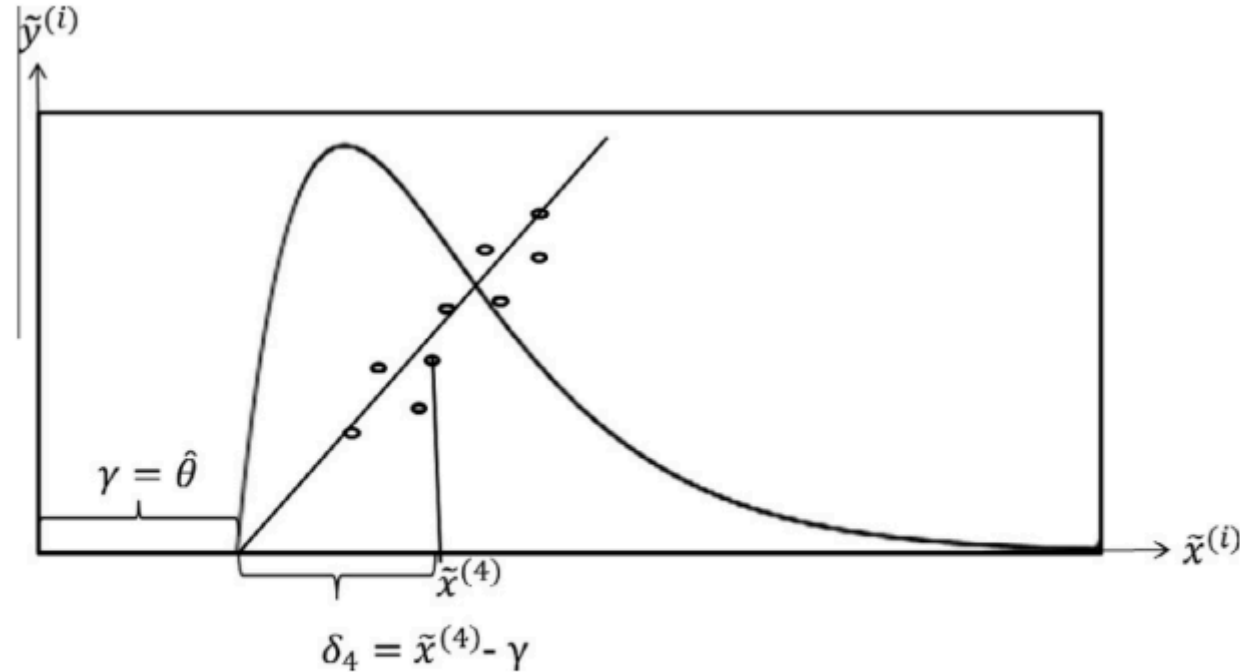


Fig. 7. Probability-plot technique for clock offset estimation. The intersection of the linear regression with the horizontal axis is taken as the minimum delay of the observed gamma distribution.

We average the end-node estimates of the offset:

$$\begin{aligned}\hat{\theta}_{A \leftrightarrow B} &= \frac{\hat{\theta}_{A \rightarrow B} - \hat{\theta}_{A \leftarrow B}}{2} \\ &= \frac{[\theta + \xi_{A \rightarrow B} - (-\theta + \xi_{A \leftarrow B})]}{2} \\ &= \theta + \frac{\xi_{A \rightarrow B} - \xi_{A \leftarrow B}}{2}\end{aligned}$$

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Estimator is unbiased when $\xi_{A \rightarrow B} \approx \xi_{A \leftarrow B}$.

Since we need the packets to be independent, we want the rate d to be such that no two packets are in a router at the same time. It is recommended to have $d \in (10, 30)$ ms.

6. Results

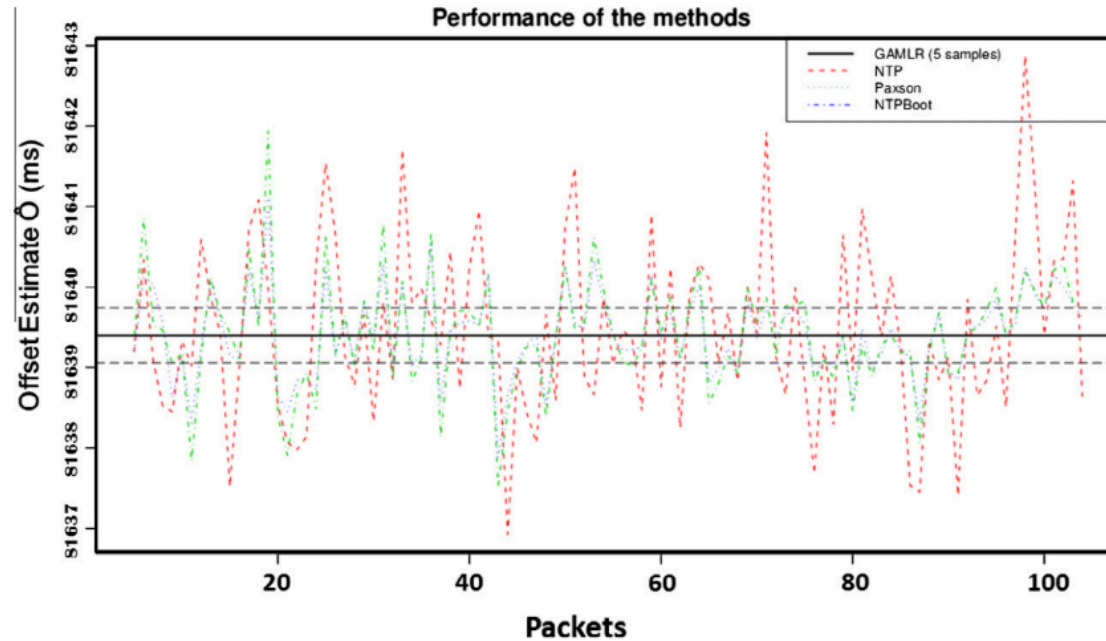


Fig. 11. Offset estimation behavior of NTP-like schemes against a sole GAMLR estimation (solid horizontal line). The dashed horizontal lines are the 95% upper and lower confidence intervals of the GAMLR estimator.

The proposed one way estimation of the queue delay is good enough to be used in robust multimedia communications protocols for ahead of time reaction before and during transmission.

E. Mota-García, R. Hasimoto-Beltran (2013), A new model-based clock-offset approximation over IP networks.